

MAYANK JOSHI

GenAI Engineer | Applied AI Engineer | AI-SDET

Ghaziabad, Uttar Pradesh | +91-8527884373 | joshimayank943@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Innovative GenAI Engineer with 4.10 years in software engineering, specializing in production-grade LLM applications, agentic AI systems, and Retrieval-Augmented Generation (RAG) pipelines. Expertise in LangChain, LangGraph, MCP, and OpenAI APIs, alongside strong skills in vector databases, hybrid retrieval, and prompt engineering for scalable AI workflows. Experienced architecting multi-agent systems and AI-powered developer tooling, deploying solutions with Python, FastAPI, and Docker.

TECHNICAL SKILLS

AI Engineering: LangChain, LangGraph, MCP, RAG, AI Agents, LLM Applications, Tool Calling

LLM Ecosystem: OpenAI APIs, Claude, Hugging Face, Ollama

Retrieval: Vector Databases, Embeddings, Semantic Search, Hybrid Search, Retrieval Optimization

AI Agents: Multi-Agent Systems, Agentic Workflows, Context Engineering, Prompt Evaluation

Engineering: Python, Java, FastAPI, Docker, SQL, TypeScript

Automation: Playwright, Selenium, GitHub Actions, Jenkins

WORK HISTORY

Senior Engineer — AI-SDET / GenAI Engineering

2025-08 – Current

Clarivate, Noida, India

- Designed and implemented agentic AI workflows using LangChain, LangGraph, and MCP, enabling LLM agents to interact with APIs, tools, and enterprise workflows.
- Created multimodal RAG workflows for document intelligence using DocLayout-YOLO, enabling extraction of text, tables, formulas, images, and captions into retrieval-ready representations with metadata traceability.
- Designed enterprise-grade RAG pipelines using embeddings, vector databases, and metadata indexing for semantic retrieval over engineering knowledge systems.
- Developed an AI-powered Allure Report Analyzer Agent that analyzes automation execution reports, identifies failure patterns, and generates root-cause insights using LLM reasoning.
- Built a natural language-driven GitHub Actions orchestration agent enabling conversational CI/CD workflow execution via LLM tool calling.
- Engineered AI-powered Playwright + TypeScript automation systems integrating LLM-driven workflows for intelligent test orchestration.
- Built AI workflow architectures using FastAPI, Docker, GitHub Actions, and vector databases for scalable AI application deployment, applying context engineering and retrieval optimization to improve observability.

Senior Software Engineer

2021-08 – 2025-08

Sopra Steria, Noida, India

- Expanded engineering scope beyond traditional automation by building AI-powered applications using Python, LLM frameworks, and retrieval-based architectures — including RAG systems, embeddings, vector databases, and semantic search pipelines for contextual knowledge retrieval.
- Developed LLM-powered applications using LangChain, OpenAI APIs, and Streamlit, including conversational AI systems and workflow automation solutions; ran multi-provider AI experimentation across OpenAI, Gemini, and HuggingFace models.
- Explored early agentic AI architectures — LLM tool calling, MCP-based integrations, and context-aware automation patterns — alongside prompt engineering, embedding optimization, and LLM evaluation workflows.
- Designed and developed scalable enterprise automation frameworks using Java, Selenium WebDriver, TestNG, Maven, and the Page Object Model, covering functional, regression, and integration testing.
- Built API automation frameworks using RestAssured for REST service and backend validation, integrated with Jenkins CI/CD pipelines for continuous regression execution and faster release validation.
- Improved automation reliability by building reusable framework components and parallel execution strategies, and optimizing reporting.

PROJECTS

Multimodal RAG System for Research Papers [GitHub](#)

- Architected an end-to-end RAG pipeline for enterprise knowledge discovery — document ingestion, intelligent chunking, embedding generation, vector indexing, and semantic retrieval — with metadata-aware and hybrid search strategies to improve retrieval accuracy.

Customer Support AI Pipeline [GitHub](#)

- Built an AI-powered customer support intelligence pipeline for transcript classification, intelligent routing, sentiment analysis, and automated quality evaluation.

MeetingMind — AI Meeting Assistant [GitHub](#)

- Built an AI meeting assistant that transforms raw transcripts into structured summaries, decisions, and actionable insights using LLM workflows.

AI-Powered Selenium Automation Code Generator [GitHub](#)

- Built a system that converts natural language test scenarios into executable Selenium/Java/TestNG automation scripts using OpenAI LLM workflows and prompt engineering, reducing manual script-authoring effort.

ML-Based Intent Classification System [GitHub](#)

- Built an NLP-based intent classification system using Universal Sentence Encoder embeddings and logistic regression for intelligent routing of automation and engineering requests.

EDUCATION

Bachelor of Technology, Computer Science

2021-06

RKGITM, Ghaziabad, India

CERTIFICATIONS

- Build AI Agents using MCP — IBM (2026)
- IBM Data Science Specialization: Python, SQL, Data Analysis & Visualization, Data Science Methodology (2026)
- Certified Tester Foundation Level (CTFL) — ISTQB (2023)

ACHIEVEMENTS

- Einstein Award — Sopra Steria (2024, 2023)
- Pinnacle Award — Sopra Steria (2024, 2023, 2022)